

Mazen Emam

Software Engineer specializing in frontend development with 4 years of experience building scalable web applications that drive business results. I help companies enhance user experiences and operational efficiency through data-driven solutions, having contributed to platforms serving manufacturing, analytics, and recruitment sectors across MENA and Europe.

Alexandria, Egypt

+20 1285-952-438

mazenemam19@proton.me

[LinkedIn/mazenemam19](#)

[Github/mazenemam19](#)

[Medium/mazenemam19](#)

[Portfolio/mazenemam19](#)

WORK EXPERIENCE

Siren Analytics, Beirut, Lebanon— *Software Engineer*

Jun 2025 - Jan 2026

Data intelligence company providing digital transformation services to the public and private sectors across Lebanon and the Middle East — **Tech Stack:** React.js, TypeScript, Material UI, React Query, REST APIs, Redux Toolkit, WebSocket, React Hook Form, Yup

- Led development of an AI chat module enabling users to query up to 300+ news sources conversationally, replacing manual content navigation across bulk-ingested datasets.
- Implemented role-based access control across a multi-tenant architecture, scoping feature and content visibility per tenant and enforcing user-level permissions across the platform.
- Engineered a multi-language translation system with React Query caching and bulk translation hooks, supporting the company's expansion into the European market.
- Pioneered a real-time WebSocket notification system using SockJS/STOMP with throttled error handling, enabling instant cross-user threat flagging that navigates recipients directly to flagged content and related articles.
- Optimized application performance by integrating React Compiler and leading application-wide code splitting, reducing unnecessary re-renders and improving bundle efficiency.
- Secured application authentication by refactoring JWT expiry logic and eliminating race conditions in token refresh, preventing unauthorized access.
- Eliminated a class of runtime errors by establishing a TypeScript enforcement pipeline with strict ESLint rules and a pre-push hook, catching type inconsistencies before code integration.

Garment IO, Giza, Egypt— *Mid-Senior Frontend Engineer*

May 2023 - Jun 2025

SaaS solution helping labor-intensive manufacturing companies enhance productivity through real-time production floor data collection — **Tech Stack:** React.js, TypeScript, Material UI, REST APIs, Redux, Sentry, Formik, Yup, AWS, Github Actions

- Introduced TypeScript and strict ESLint enforcement across a 6-year-old codebase spanning 10+ modules, and 40+ pages through an incremental migration strategy, establishing type safety without disrupting active development.
- Modernized the build pipeline by migrating from Create React App to Vite, reducing production build time by 50% and improving local development speed by 5x.
- Designed an i18n system with real-time language switching across 4 languages, enabling product deployment across MENA and European markets.
- Implemented Sentry performance monitoring capturing Web Vitals, API latency bottlenecks, identifying production issues across 30+ manufacturing facilities.

TECHNOLOGIES

Proficient

React.js,

TypeScript, JavaScript,

Redux/Redux Toolkit,

React Query/TanStack Query,

Material-UI, Vite, HTML, CSS,

SASS, Bootstrap, Git

Familiar

Next.js, Node.js, Express,

MongoDB, GraphQL,

WebSocket/STOMP, MobX,

Sentry, Jira, Jest, Vitest, React

Testing Library, AWS, Docker,

Webpack

Education

Udacity

[React Nanodegree](#)

Jul 2022 - Aug 2022

Free Code Camp

[JavaScript Algorithms And Data Structures](#)

Jun 2022 - Jul 2022

Information Technology Institute

[MEARN Track, 3 Months diploma](#)

Sep 2020 - Dec 2020

Alexandria University

BSc. in Management

Information Systems, Faculty of Commerce

Oct 2013 - Jul 2017

- Built a QR code scanning system for line managers, embedding machine IDs into scannable codes and eliminating manual machine lookup across 30+ manufacturing facilities.
- Resolved performance bottlenecks and memory leaks by converting sequential API calls to parallel execution, removing unused data fetching, and enforcing pagination on unbounded endpoints across data-heavy views.

LANGUAGES

Arabic Native

English Fluent

[AsgaTech](#), Giza, Egypt— *Frontend Engineer*

Nov 2022 - May 2023

Custom software firm with 19+ years serving clients across MENA — **Tech Stack:** React.js, TypeScript, Material UI, REST APIs, MobX, React Hook Form, Yup

- Established a reusable component library with RTL support, creating a consistent design system that reduced new feature development time by 50%.
- Executed full application internationalization with dynamic RTL switching, enabling product launch in Arabic and English markets.
- Accelerated app startup speed by 35% through image optimization, lazy loading, and efficient API integrations.
- Developed a multi-step registration form using React Context, submitting mandatory fields incrementally per step rather than in bulk — reducing backend load from incomplete submissions while streamlining the onboarding experience.

[Akwad](#), Alexandria, Egypt— *Frontend Engineer*

Oct 2021 - Jul 2022

Custom technology solutions and component library development company.

Tech Stack: React.js, JavaScript, Bootstrap, REST APIs, Context API, React Hook Form

- Created a reusable form and grid component framework, standardizing development and reducing page build time by 90%.
- Delivered a complex MasterDetail system for asset and order management, enabling critical business workflows.
- Overhauled the core data grid infrastructure used across the majority of the system's forms and views, unifying pagination and search logic to eliminate inconsistent behavior and reduce bugs.
- Replaced silent failures with a centralized error handling system featuring user-facing toast notifications and admin oversight logs, enhancing debuggability.

PROJECTS

Job Radar [\[Github\]](#) | [\[Live Demo\]](#).

Personalized AI-powered job dashboard for Senior React/Next.js roles (Egypt-based profile) — **Tech Stack:** Next.js, TypeScript, Tailwind, Supabase, Gemini API

- Built AI filtering UI (Regex + Culture Detector + Gemini) cutting job noise by ~85%, surfacing high-match Senior React/Next.js roles with transparent decisions.
- Architected modular Next.js frontend with centralized constants/types, standardizing scoring across pipelines and cutting feature dev time by ~40%.
- Designed seniority-aware routing dynamically adjusting gates per pipeline (Strict Local, Mid+ Global), boosting opportunities to find international jobs.
- Built a Market Intelligence dashboard with trend analysis, skill co-occurrence charts, and missing-skill using React and Supabase.
- Implemented production safeguards via CRON_SECRET validation, secret isolation, and optimistic UI updates—securing workflows without sacrificing UX.